

Design Choices for Embodied Conversational Agents to Preserve Testimonies by Contemporary Witnesses

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ABSTRACT

Conversations with contemporary witnesses, e.g., Holocaust survivors, are valuable opportunities to learn about impacts of historical events. They lead to a more thorough conveyance of history by complementing aggregated facts and numbers with emotional personal experiences and detailed accounts. Encounters and interviews with first-hand witnesses to a number of historical periods soon are a thing of the past due to the respective survivors' advanced age. Embodied Conversational Agents using recordings of contemporary witnesses are an attempt at preserving interactive and personal testimonies for future generations of students and educators. Since the implementation of such Interactive Digital Testimonies (IDT) contains irreversible design decisions, an overview of available choices and their respective consequences are needed. While a number of IDTs have been created over the past years, there is little empirical research on available design choices and the resulting effects on users. In my dissertation I will identify the required HCI features of IDTs, as well as investigate and evaluate how these features can be implemented. This scientific basis will allow for informed decisions during the development of future IDTs.

CCS CONCEPTS

• **Human-centered computing** → **User studies**; Natural language interfaces; *Displays and imagers*; *Virtual reality*; **User centered design**; *Empirical studies in HCI*; • **Applied computing** → **Interactive learning environments**.

KEYWORDS

Embodied Conversational Agents; Interactive Digital Testimony; Learning; Modality; Presence; Immersion; Accessibility

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1 RESEARCH SITUATION

I am fourth-year Ph.D. student of computer science at LMU Munich, advised by Dieter Kranzlmüller. As researcher at the Centre for Virtual Reality and Visualisation (V2C) of the Leibniz Supercomputing Centre (LRZ), I facilitate the design, implementation, and evaluation of several Interactive Digital Testimonies (IDTs). My work also includes Virtual Reality visualizations of supercomputer simulations of climatological data, using Augmented Reality to raise awareness for endangered species, and supporting Extended Reality initiatives. With a number of empirical studies and literature reviews finished, I intend to complete my thesis in 2023. My research complements my enthusiasm for history education and my desire to impede historical revisionism.

2 CONTEXT AND MOTIVATION

Oral history and interviews with contemporary witnesses are affective and engaging means of education [30]. Unlike abstract numbers and reports, encounters with survivors of historical events can provide immediate and tangible access to experiences. Combined with further context and factual data, they can become a powerful pedagogical tool [7]. A current, transnational, and prevalent use case of oral history is Holocaust education [5, 6, 24]. However, more than 75 years after the end of World War II the number of remaining survivors of the Holocaust is dwindling.

In preparation for the end of “The era of the witness” [31], several projects, e.g., the USC Shoah Foundation, the Forever Project, and LediZ (“Lernen mit digitalen Zeugnissen”, transl. “Learning with digital testimonies”), designed and implemented Interactive Digital Testimonies (IDTs) to preserve interactive conversations with Holocaust survivors for future generations (see Figure 1). Building on advances in machine learning as well as display and capturing technology, IDTs combine natural language processing (NLP) systems with prior recordings of answers of witnesses. This allows users to interact with the virtual replication of witnesses by posing questions, which are analyzed and met with the most suitable pre-recorded responses. With diverse modalities for input (text, speech) and output (text, speech, monoscopic, stereoscopic, volumetric), IDTs have a range of implementation requirements as well as varied effects on users.

While the production of IDTs is ongoing, there is little empirical research on their design choices and the resulting effects on users. Although IDTs can be understood as non-synthetic Embodied Conversational Agents (ECA) and build on available research, the benefits and limits of using prerecorded responses require targeted examination. Additionally, issues like accessibility rarely receive



Figure 1: A user interacting with the IDT of Holocaust survivor Abba Naor, created by the project “LediZ”. The smartphone serves as voice input device for questions. The system analyzes the intent of the question and displays the best fitting prerecorded audio-visual response in stereoscopic 3D.

the necessary attention during the design and implementation processes. Since the recording process is stressful for the witness, time-consuming, and costly, design mistakes can be difficult or even impossible to remedy.

My work aims to close this research gap by investigating and evaluating the human-computer interaction aspects of IDTs. It will provide a thorough overview of available features and their respective impacts, which is needed to avoid the aforementioned design problems.

3 RELATED WORK

An early IDT precursor was installed in the Nixon presidential library in 1990 [8]. Visitors could “Ask the President” by selecting one of 280 questions, which signaled the display to show a recording of Richard Nixon’s corresponding answer. However, the response videos were taken from different sources and, therefore, audio-visually non-uniform. Users also were not able to pose their own questions. In 1998, “Synthetic Interviews” combined speech recognition software with purpose-made recordings of actors portraying famous historic individuals [19]. The concept also included recordings of the characters idling around between questions as well as fallback answers if no fitting response was available. Due to the use of actors, the content, actions, and reactions in the recordings are not authentic with regard to the depicted person. Building on the same concept, “Ben Franklin’s Ghost” used the Pepper’s ghost illusion for added realism in 2005 [14].

Beginning in 2014, use cases in Holocaust education caused the most significant conceptual and scientific progress for IDTs so far [3, 4, 16]. These current designs share a number of characteristics [2, 15, 17, 29]:

- Purpose-made audio-visual stereoscopic 3D recordings of contemporary witnesses detailing their own life experiences
- Implementation-dependent modality options for input and output
- Minimalistic visual settings and backgrounds
- More than 1000 prompts, including introductory witness accounts, idle loops, and default fallback responses
- Stationary as well as web-based options

A drawback common to all IDTs is the strict limitation in content, stemming from the constraint of using only prerecorded replies. As responses can not be synthesized, there will always remain questions without matching answer. Since the IDT is unable to reference arbitrary prior exchanges, limits the depth of the conversation between user and computer to a quasi-interactive level [23]. Similarly, responses can not adapt to topical issues or geopolitical changes. For example, IDTs created before 2020 can not be questioned about the SARS-CoV-2 pandemic. The erroneous impression of an IDT being able to have the same conversational abilities as a real human represents a possible false affordance [21], which can decrease user acceptance.

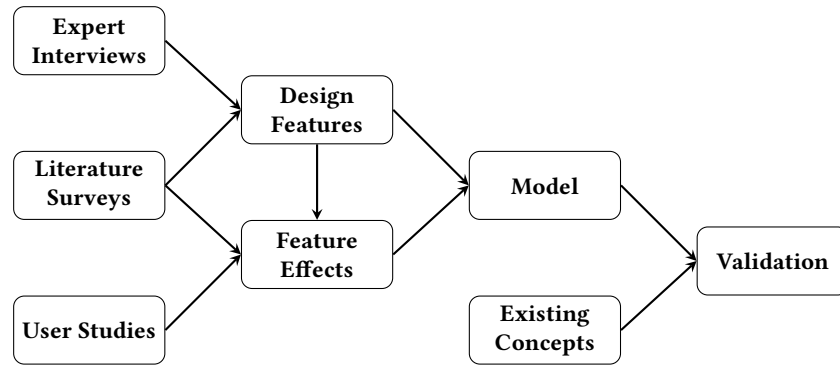


Figure 2: Proposed research process.

Realism of humanoid ECAs is important for many aspects of user interaction, like presence, involvement, enjoyment, and credibility [1]. Select requirements for this realism are multi-modality in in- and output, fidelity of presentation, personality, consistent behavior [12], and social cues [10]. Artificially (re-)creating humans virtually is, therefore, complex and challenging. It requires a realistic visual appearance [28], a matching voice [20, 27], and intricate behavior models [25] to avoid the Uncanny Valley Effect [18, 22].

There are to date few published studies investigating the effects of IDTs, covering engagement [29], co-presence [2], and realism [15]. An upcoming approach is the use of volumetric video data [9], which also needs to be evaluated and compared to existing concepts, including conversations with real contemporary witnesses.

4 RESEARCH QUESTIONS

In my thesis I will answer the following research question:

How can interactive and immersive systems using multimodal displays and supported by NLP be designed to authentically replicate conversations with real humans using prerecorded responses?

I intend to (1) create a conceptual model which lists the relevant design features, their effects, and their interdependencies. I will then (2) evaluate and validate the model by mapping existing IDT concepts on the components of the model. Since I strictly focus on the human-computer-interaction aspects, didactics (e.g., quality of learning), machine learning (e.g., quality of matching responses), or ethical considerations (e.g., consent for continued use of the witness' testimony after their passing) are not within the scope of my research.

5 RESEARCH APPROACH

I structure my work in four cascading steps (see Figure 2). First I gather and refine relevant *Design Features* by means of *Literature Surveys* and semi-structured *Expert Interviews*. As suitable experts for my research I consider specialists with vast experiences with real and digital witness interviews, e.g., educators in school or museum settings, designers or creators of IDTs, and contemporary witnesses themselves. Using Grounded Theory [11], I organize and

interpret the resulting qualitative data. The emerging *Design Features* serve as nodes in the *Model*. Subsequently, I add relationships (*Feature Effects*) between these nodes, based on further *Literature Surveys* and quantitative data from my *User Studies*. I focus on comparative studies investigating the consequences of varied design choices on user perception. Exemplified relationships are implications, preconditions, interdependencies, or correlations. In the final step, *Validation*, I compile an exhaustive list of *Existing Concepts* of IDTs, map their design choices onto the *Model* and compare the predicted characteristics with the real effects.

6 PRELIMINARY RESULTS

By facilitating and accompanying the creation of multiple IDTs over the past three years, I have gained valuable insights as well as access to prototypes and IDTs with modifiable components. I have been attending to content selection, recording of contemporary witnesses, implementation, video hosting, hardware provisioning, along with field and laboratory studies. The design processes are accompanied by extensive literature surveys on ECAs, oral history, and research methods in general. Simultaneously, I have kept in touch with Holocaust survivors, history teachers, museum guides, as well as international researchers of history education, civic education, and conversational agents.

Unfortunately, the SARS-CoV-2 pandemic and the necessary accompanying restrictions have inhibited planned empirical studies and slowed my overall research progress. I counteracted these issues by capitalizing on temporary liftings of restrictions and supporting the creation of online IDTs, which I could then use to offer socially distant and location-independent participation in user studies online. However, these online studies allow for less control over environmental variables and interferences and do not qualify for all intended experimental settings.

In an early study [15] I have found first empirical results, like the significance of a well-rounded response matching system, the impact of introductory testimonies, the challenges for location-independent IDT implementations, and the overall serviceability of the chosen interaction design. In a second, comparative study I have identified the modality-based differences in user experience between audio-only, audio-visual 2D, and audio-visual stereoscopic 3D IDTs. I found all three modalities to be viable, however, the additional visual representation of the virtual witness increased user

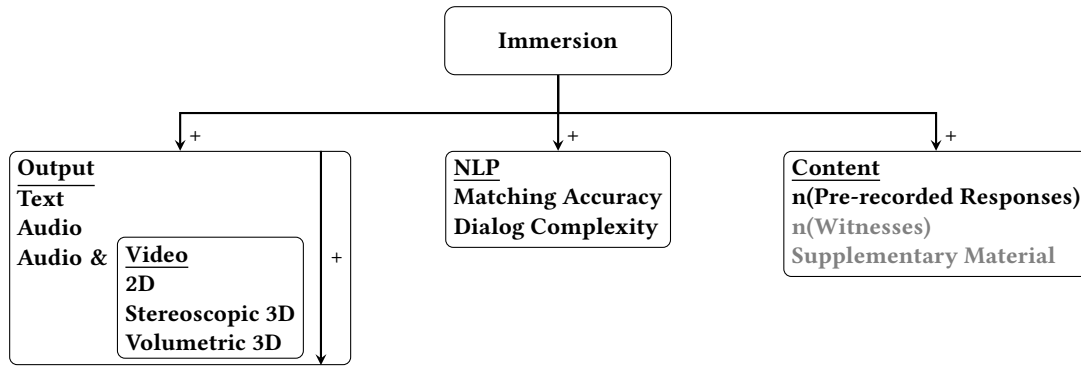


Figure 3: Excerpt of the preliminary model of design choices. It lists IDT characteristics which influence the perceived immersion during usage. The increased immersion of volumetric 3D IDTs is a hypothesis which I will test in an upcoming experiment.

experience and presence. Stereoscopic 3D IDTs were perceived as more realistic than 2D IDTs, but the increased discomfort caused by the implementation of stereoscopic viewing limited increases in perceived presence and emotionality. The corresponding submission for publication is currently under review.

Figure 3 shows an excerpt of the preliminary model, based on my findings. The level of immersion of an IDT correlates with the perceived realism of the contemporary witness. Lifelike 3D display with authentic conversational abilities lead to a persuasive presentations. Lackluster responses or uncanny displays break this immersion. Adding further design objectives, like accessibility, leads to complex relationships between design features. They require the provisioning of flexible modality combinations, since, e.g., stereoblind users do not profit from stereoscopic displays, whereas users with hearing impairments require additional visual data, like subtitles or signed languages.

My next steps are the methodical preparation, validation, and execution of expert interviews, which will uncover further design choices, perspectives, and relationships. I will then conduct quantitative user studies using within-subjects design to measure and compare not-yet investigated design choices, like the differences between stereoscopic 3D and volumetric 3D IDTs. I intend to use ANOVA to analyze the resulting data.

7 EXPECTED CONTRIBUTIONS

My research will make the following contributions:

- A checklist of IDT design choices
- A model of relationships and interdependencies of IDT design choices
- Findings on lifelike ECAs and non-synthetic virtual humans

In the long term, my research will help guide future IDT concepts and implementations for various use cases digitally preserving conversations with real humans. These use cases are not limited to educational contexts and can include awareness campaigns [13], interactions with celebrities, or private contexts [26], e.g., conversations with one's ancestors.

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